

Quinn Asena

Ecologist & Data Scientist

Cary Institute of Ecosystem Studies

Forest Futures Lab, New York State

CONTACT

✉ asenaq@caryinstitute.org

📧 quinnasena.github.io

🌐 github.com/QuinnAsena

🆔 ORCID 0000-0002-4086-460X

in LinkedIn

LAB AFFILIATIONS

Forest Futures Lab

Cary Institute

Williams Paleoecology

UW-Madison

Perry Lab

U. Auckland

Neotoma Paleoecology DB

Leadership Council

SKILLS & TOOLS

R · Python · c++ · Bash · HPC
· Git

INTERESTS

Classical piano · Rock climbing
· Art

RESEARCH INTERESTS

I'm interested in ecological dynamics and mechanisms that underlie change in ecosystems over time and space. My research focuses on biotic interactions and how ecosystems respond to environmental and climate change. Much of my previous research has focused on long-term ecological trajectories and palaeoecology. I am now exploring contemporary ecosystem change through process-based models.

PROFESSIONAL EXPERIENCE

Data Scientist

Cary Institute of Ecosystem Studies

New York State

2025 – Present

- Modelling boreal forest dynamics and ecological processes under climate change and fire regimes.

Postdoctoral Research Associate

University of Wisconsin-Madison

Madison, WI

2022 – 2024

- NSF-funded project developing state-space modelling methods to analyse palaeoecological records.
- Abrupt change in ecosystems project led by Jack Williams and Anthony Ives.

Engagement Specialist

Centre for e-Research - University of Auckland

Auckland, NZ

2021 – 2022

- Provided research-computing support for graduate students and research staff: virtual machines, data storage, and analytical workflows.
- Technical skills: Linux, Bash, Slurm, HPC, and version control.

Data Analyst

Auckland Council

Auckland, NZ

2021

- Analysed citizen science bird count data and produced a technical report.

RESEARCH EXPERIENCE

Research Assistant

University of Auckland

Auckland, NZ

2021 – 2022

- Conducting a bibliometric analysis on the topic of climate justice.

Research Assistant

Stockholm Environment Institute

York, UK

2016 – 2017

- Reconstruction of fire regimes in the North York Moors.
- Microscope analysis and core preparation for charcoal and spheroidal carbonaceous particles.
- Statistical analysis and compilation of data.

TEACHING EXPERIENCE

PhD Student Mentor

SUNY

Virtual

2025

- One-on-one mentoring in statistics for palaeoecology.

Graduate Teaching Assistant & Assistant Coordinator

University of Auckland

Auckland, NZ

2018 – 2020

- Assistant coordinator for Discovering Environmental Modelling.
- Guest lecturer on population growth. Developed a [Shiny application](#) as a lecture aid.
- GTA for: Natural and Human Environmental Systems; Discovering Environmental Modelling; Environmental Science and Management.

EDUCATION

PhD, University of Auckland

School of Environment

Auckland, NZ

2017 – 2021

- Exploring virtual ecological methods for generating pseudoproxy data to assess statistical inferences under data uncertainty.
- Supervised by George Perry and Janet Wilmschurst.

MEnv. Environmental Science

University of York

York, UK

2012 – 2016

- First-class integrated master's degree in environmental science. Master's research involved environmental monitoring using sensor networks and robotics.
- Courses included: Ecological Principles; Biogeography; Applied Ecology and Environmental Management; Climate Change: Science, Observation & Impacts; Research Skills and Statistical Methods.

Geocomputation & Machine Learning for Environmental Applications

Basilicata University

Matera, Italy

2025

- Certified course in machine learning for spatial analyses.

Certified Carpentries Instructor

The Carpentries

Global (Virtual)

2023

- Certified to teach and host [Carpentries](#) workshops in data science and computational skills.

Performance Composition & Theory

Royal Academy of Music

London, UK

2005 – 2010

- Study in performance, composition, and musical theory. Multiple competition awards.
- ABRSM Grade 8 Piano Performance.
- ABRSM Grade 5 Music Theory.

PUBLICATIONS

- **Asena, Q.**, Williams, J., Stefanova, V., Johnson, J., Shuman, B., Ives, A. (2026). Statistical analyses of ecological multinomial time series to identify environmental drivers and biotic interactions. *Methods in Ecology and Evolution*.
- **Asena, Q.**, Perry, G.L., Wilmschurst, J. (2026). Information loss in palaeoecological data from process and observer error. *Climate of the Past*.

- Lee, F., Bellve, A.M., Alena, H., **Asena, Q.**, Perry, G.L. (2025). [Using population viability analysis and fossil records to inform the conservation of pateke \(*Anas chlorotis*\)](#). *New Zealand Journal of Ecology*.
- **Asena, Q.**, Perry, G.L., Wilmshurst, J. (2024). [Is the past recoverable from the data? Pseudoproxy modelling of uncertainties in palaeoecological data](#). *The Holocene*.
- Parsons, M., **Asena, Q.**, Johnson, D., Nalau, J. (2024). [A Bibliometric Review and Topic Analysis of Climate Justice Literature: Mapping Trends, Voices, and the Way Forward](#). *Climate Risk Management*. Code repository publicly available.
- Courtney-Mustaphi, C.J., Brahney, J., Aquino-Lopez, M.A., Goring, S., Orton, K., Noronha, A., Czaplewski, J., **Asena, Q.**, Paton, S. (2019). [Guidelines for Reporting and Archiving 210Pb Sediment Chronologies to Improve Fidelity and Extend Data Lifecycle](#). *Quaternary Geochronology*.
- Heinemeyer, A., Burn, W.L., **Asena, Q.**, Jones, A., Ashby, M. (2019). [Response to comment on Peatland carbon stocks and burn history](#). *GEO: Geography and Environment*.
- Heinemeyer, A., **Asena, Q.**, Burn, W.L., Jones, A. (2018). [Peatland carbon stocks and burn history: Blanket bog peat core evidence highlights charcoal impacts on peat physical properties and long-term carbon storage](#). *GEO: Geography and Environment*.
- Simpkins, C.E., Lee, F., Powers, B.F., Anderson, S., **Asena, Q.**, Brock, J.M.R., Perry, G.L. (2018). [Population viability analyses in New Zealand: a review](#). *New Zealand Journal of Ecology*.
- Hamilton, R., Brussel, T., **Asena, Q.**, Bruel, R., Marcisz, K., Slowinski, M., Morris, J. (2018). [Assessing the links between resilience disturbance and functional traits in paleoecological datasets](#). *Past Global Changes Magazine*, vol. 26(2), 87.

SOFTWARE & OPEN SCIENCE

multinomialTS

R package

- R package for state-space modelling of multinomially distributed ecological time-series data.

State-Space Methods for Paleoecological Hypothesis Testing

Workshop materials

- Open workshop materials for state-space modelling of community and palaeoecological time series.

Authoring Collaborative Research Projects in Quarto

Workshop materials

- Resources for collaborative research workflows using Git and GitHub and Quarto (ResBaz 2022–2024).

fisherR

R package

- R package for calculating Fisher's Information on ecological time-series data.

Population Growth Explorer

Shiny app

- Interactive app exploring population growth equations for undergraduate teaching.

CONFERENCES, TALKS & WORKSHOPS

Neotoma Outreach

2026

Virtual

- Talk: [A Virtual Ecological Approach to Modeling Uncertainties in Paleoecology](#)
- Workshop: [Hands-on workshop for age-depth modelling using Behron](#)

AGU - American Geophysical Union

2025

New Orleans, LA

- Poster: Assessing climate-driven compositional shifts in Boreal Forests using iLand.

- ESA 2024 - Ecological Society of America** 2024
Long Beach, CA
- **Speaker:** Hypothesis testing of the drivers of long-term ecological change in Michigan.
 - **Workshop host:** [State-Space Methods for Testing Hypotheses About Environmental Drivers and Biotic Interactions](#)
- INQUA 2023 - International Union for Quaternary Research** 2023
Rome, Italy
- **Lead Convenor:** [Data science and paleoecology: current intersections and advances](#)
- ESA 2023 - Ecological Society of America** 2023
Portland, OR
- **Speaker:** [Modelling palaeoecological community data: a state-space approach](#)
- ResBaz - Research Bazaar** 2021 – 2024
Auckland, NZ; Madison, WI
- Organiser: ResBaz Aotearoa 2021.
 - Workshop lead: [Quarto for Collaboration](#) at ResBaz NZ 2022–2024 and in-person Madison, WI, 2023.
- AGU - American Geophysical Union** 2022
Chicago, IL
- Speaker: virtual ecological methods for assessing uncertainty in palaeoecological data using pseudo-proxies.
- AMQUA - American Quaternary Association** 2022
Madison, WI
- Poster: virtual ecological methods for assessing uncertainty in palaeoecological data using pseudo-proxies.
- BES Palaeo in R - British Ecological Society** 2022
Virtual
- Speaker: virtual ecological methods for assessing uncertainty in palaeoecological data using pseudo-proxies.
- INQUA 2019 - International Union for Quaternary Research** 2019
Dublin, Ireland
- Speaker: pseudoproxy modelling for assessing statistical inferences.
- Durham University - Invited Speaker** 2019
Durham, UK
- Modelling resilience in ecosystems.
- Biological Heritage Science Challenge** *Christchurch, NZ*
2017 – 2019
University of Canterbury
- Speaker on resilience in ecosystems.
- International Swiss Climate Summer School** *Bern, Switzerland*
2018
University of Bern
- Poster presentation and participant. Accredited course (3 ECTS credits).
- EcoRe3: Resistance Recovery and Resilience in Long-term Ecological Systems** *Utah, USA*
2018
PAGES - University of Utah

- Speaker: modelling resilience in ecosystems.

Te Punaha Matatini Biomaths

University of Auckland

Auckland, NZ

2018

- Speaker.

PEER REVIEW

Ecology Letters

Journal of Open Source Software

Nature Communications

Environmental Research and Technology

Quaternary Geochronology

Methods in Ecology and Evolution

SERVICE & LEADERSHIP

Early Career Representative

Neotoma Leadership Council

Virtual

2024 – Present

- Early career representative on the Neotoma leadership council.